VESIT Climate Action Roadmap 2024-2040: Net-Zero Trajectory Modeling

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Authoring Body: Campus Decarbonization Working Group and Department of Artificial Intelligence & Data Science (DEP_AIDS).

Institutional Target: TRG_NET_ZERO (Net-Zero GHG Emissions by December 31, 2040)

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. STATUTORY CHARTER AND BASELINE REFERENCE

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) in Chembur, Mumbai, adopts this binding Climate Action Roadmap to systematically steer institutional capital investments, facilities modernization, and operational policies toward absolute net-zero operational greenhouse gas (GHG) emissions by December 31, 2040. The roadmap encompasses all academic and administrative facilities across 32,500 m2 of built infrastructure.

The comprehensive calendar year 2024 baseline audit (DOC_AUD_002) established the verified institutional carbon footprint:

- Total Baseline Emissions (E0): 2,450,000 kg CO2e/year (2,450 metric tonnes CO2e).

- Scope 1 Direct Emissions: 49,312 kg CO2e (2.01%), driven by 18,400 L of diesel combustion in emergency generators (EF_DIESEL_GEN = 2.68 kg CO2e/L).

- Scope 2 Indirect Imported Electricity Emissions: 1,681,000 kg CO2e (68.61%), resulting from 2,050,000 kWh of grid electricity purchase (EF_ELEC_GRID = 0.82 kg CO2e/kWh).

- Scope 3 Commuting, Waste, and Water Emissions: 719,688 kg CO2e (29.38%), comprising commuter transit (580,000 kg CO2e), municipal landfill waste (115,000 kg CO2e), and water processing (24,688 kg CO2e).

2. MATHEMATICAL DECARBONIZATION TRAJECTORY MODELING

To achieve net zero over the 16-year operational horizon (t0 = 2024 to t_target = 2040), the Department of AI & Data Science (DEP_AIDS) developed two mathematical trajectories:

2.1 Model A: Linear Decarbonization Trajectory (Constant Annual Abatement)

E_linear(t) = E0 - r * (t - t0)

Where:

- E0 = 2,450,000 kg CO2e

- Planning Horizon = 16 years

- Required Annual Linear Abatement Rate: r = 2,450,000 / 16 = 153,125 kg CO2e/year.

Under this trajectory, emissions must decrease to:

- 1,990,625 kg CO2e by 2027

- 1,531,250 kg CO2e by 2030

- 765,625 kg CO2e by 2035

- Exactly 0 kg CO2e by December 31, 2040.

2.2 Model B: Compound Annual Reduction Trajectory (Exponential Decay)

E_compound(t) = E0 * (1 - p)^(t - t0)

Requiring residual emissions by 2040 to remain below 2.0% of baseline (< 49,000 kg CO2e), the required compound abatement rate is p = 22.8% per annum if relying strictly on operational abatement, or p = 12.5% per annum when coupled with high-integrity local biological sequestration (INIT_BIODIVERSITY_ZONE).

The Taskforce adopts Model A (Linear Decarbonization) as the statutory institutional compliance benchmark, reviewed quarterly on the EcoClock AI interactive platform.

3. THREE-PHASE IMPLEMENTATION MILESTONES

- Phase I (Immediate Execution, 2024 - 2027): Focuses on low-hanging energy efficiency and on-site generation. Commission Phase-2 Solar (INIT_SOLAR_PH2, 200 kWp + 100 kWh BESS) across parking lots and BLD_AUD, cutting 232,880 kg CO2e. Expand INIT_ELECTRIC_SHUTTLE to four electric vans, abating 95,000 kg CO2e. Enforce IoT thermostat lockdowns (INIT_IOT_ENERGY_MGT), saving 80,360 kg CO2e. Full canteen food waste biomethanation (INIT_BIOGAS_CANTEEN), abating 48,000 kg CO2e. Phase 1 cumulative annual abatement: 456,240 kg CO2e/year.

- Phase II (Deep Institutional Transformation, 2028 - 2032): Execute green power wheeling agreements (1.2 MW off-site group captive solar-wind farm under open access regulations) to displace remaining imported grid electricity, eliminating 1,150,000 kg CO2e. Expand rainwater harvesting cisterns to 650,000 L and mandate fleet electrification for all contracted campus staff buses.

- Phase III (Residual Decarbonization and Sequestration, 2033 - 2040): Transition all HVAC chillers to ultra-low GWP refrigerants (R-1234ze / R-290). Mature biomass growth across the expanded Miyawaki forest (INIT_BIODIVERSITY_ZONE) sequesters 38,000 kg CO2e/year, neutralizing residual operational emissions to achieve absolute Net-Zero by December 31, 2040.

4. MONITORING AND VERIFICATION GOVERNANCE

Progress shall be continuously tracked using the EcoClock AI GraphRAG analytics engine, connecting utility telemetry directly to Neo4j knowledge graphs to guarantee evidence-grounded accountability advancing United Nations SDG 7, SDG 11, SDG 12, and SDG 13.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_CLM_017**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*